CSC 3320 Final Project Proposal

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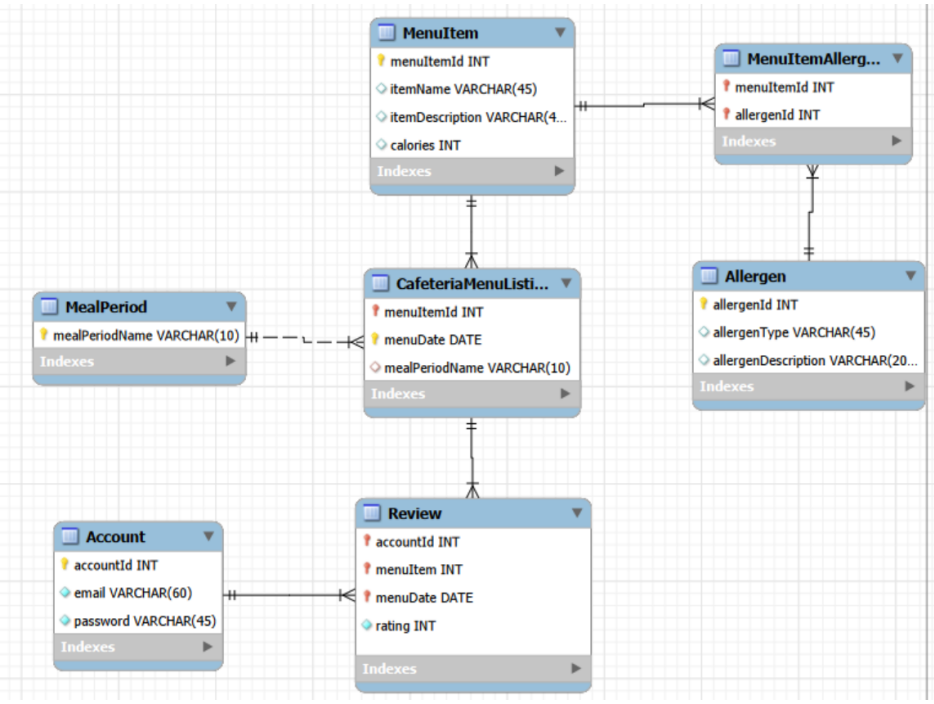
Inspired by a project I am working on in my Software Requirements and Architecture class, I wish to design a database revolving around a cafeteria forum and feedback application. There is a problem here on MSOE campus that the Campus Center cafeteria food has variable quality, and students have no way to give feedback on it. This database would consist of user reviews tied to food items, which can be used to inform future menu changes.

The intended users of this database are MSOE students (or other people who pay for dining services) who input reviews into the system, and Cafeteria Staff who will benefit from queries from this database to recognize trends in user opinion and decide on what menu items will be featured in the cafeteria in the future.

The primary workflow of the student is that they log into the application as a user, go to a cafeteria menu from a given day from a user interface, select a food item, then leave a 1–5-star review.

The primary workflow of a Cafeteria Staff member would require they be in a separate interface (or even working with MySQL itself). They would select a start date and an end date and generate a report seeing the ratings and average ratings of all menu items from those dates.

The Minimum Viable Product would be a website which modifies SQL table entries in a database on the same machine. A stretch goal of this project would be to divide it into two applications: a client-side website, and a server-side database that do not exist in the same folder.



Business rules:

- Menu items may contain 0:N allergens to follow laws documenting food allergens.

Assumptions:

- Items may be categorized under a meal period (breakfast, lunch, dinner, etc.).

Constraints:

- Accounts are anonymous to dissuade social anxiety from using the platform.
- Reviews are limited to 5-star ratings. Therefore, a valid rating must be between 1-5.

Other Major Decisions:

- Accounts rate menu items based on date. This is so they can rate menu items again in the future if the quality of the item changes.

Aside from SQL, the technology stack would include HTML, CSS, and JavaScript for the front-end for students. This would fit because students want a user-friendly way to interact with the system, and the system designers do not want unauthorized users to tamper with the database files directly.

Students can only upload reviews through buttons (a submit button and 5 star icons). We don't plan to include text input, but we should try to use parameterized queries when using these elements to prevent users from tampering with the database using SQL code in a strange spot.

A potential “create” query would be to take an existing list of Allergen names from a file and add them to the dataset. There will also need to be creation queries to represent creating accounts or uploading reviews. An advanced “read” query would be to display the average rating for each menu item served on a specific date during Lunch period. A potential “update” query would be to increase the calories of every menu item that includes the allergen “Soy” by 5 unless it also contains “Dairy”. An idea for a “deletion” query would be that a specific Account wants to delete all reviews it posted for menu items under 40 calories.

I anticipate using AI as an advisor to ensure the current database structure and project meets weekly goals. AI could also be used as a “white hat” hacker to expose any failures in the security of my database or website. To prevent leaking information, no information regarding MSOE or myself will be shared. Sensitive references will be replaced with dummy values and names.

Starting-Point Reflection:

I feel confident in writing the database, and I am confident in writing a website. What I am not sure of yet is how (or if I can) use a website to interact with an SQL database. The biggest challenge will be using SQL prompts in JavaScript code, and ensuring users cannot inject their own SQL code. Otherwise, I have a good idea of what tables can be logically added to this database.

I use AI occasionally, meaning I know how to write a good prompt, and have used AI either to double check my work or explain complex topics to me (especially with unfamiliar programming languages). To verify the AI output is correct, I will refer to my own notes and previous experience from this course to make sure any database advice is sensible. When it comes to advice about website integration, unit (or manual) testing of SQL commands will be needed to ensure the advice is correct.

I want to learn how to apply my knowledge of database creation to a real-world application that can theoretically be used by a company. To take the project to the next step, I’d need to learn how to give a client-side only user information and have them send commands to a server-side only application. That might require a server like Amazon Web Services Database to host the main database, and permissions for each user to upload to that database. That raises another concern of how to prevent users from posting unauthorized/unexpected material to the database, but that is currently outside the scope of the project.